# GUARDIAN / CONVERGENCE — CURRENT ARCHITECTURE

NetClaw host 192.168.3.252 • Docker Compose + systemd • Verified 24 Oct 2026



OPERATOR ACCESS

**Operator browser**
Chat · Convergence · Canvas
`Remote subnet → 192.168.3.252:3000`

Same origin

**New HUD UI**
Vite preview · host TCP :3000
`Convergence is inline; no legacy iframe`

/api/*

**HUD API + Convergence proxy**
Loopback 127.0.0.1:3001
`/api/home/* · /api/models · /api/chat`

MANAGED ENVIRONMENT

**Network devices + WAN targets**
pfSense · UniFi APs · campus switches
Device exporters / APIs / logs / reachability probes

01 SNMP / syslog / UniFi / WAN probes

## OBSERVABILITY + CASE DATA

Docker Compose: netclaw-convergence

**Collectors, exporters and probes**
OTel: switch SNMP + device syslog TCP/UDP :1514
Blackbox :9115 · UniFi :9899 · speedtest / Pushgateway
OTel fans metrics to Prometheus / VictoriaMetrics

Metrics

**Metrics stores**
Prometheus :9090
VictoriaMetrics :8428
PromQL / time series

Logs

**Log stores**
Loki :3100
VictoriaLogs :9428
Syslog / device evidence

Alert rules

**Alertmanager**
TCP :9093 · groups / deduplicates firing alerts
`Webhook → host alert-receiver :8099`

02 POST /webhook

Alert state

**Convergence API**
Host :3080 → container :3000
Health / Wi-Fi / devices / alerts / diary / triage
Case identity: site + event_id + alert fingerprint
No LLM runs here; telemetry and case persistence

SQL case lifecycle

**PostgreSQL**
Internal Docker :5432
Events / notes / feedback

**Grafana**
Host :3300 → :3000
Queries metrics / logs

PromQL reads

Metrics / logs

Convergence /api/home/* proxy; bearer added server-side

## NETCLAW CONTROL SERVICES

Host processes / systemd; existing NetClaw installation

**Alert receiver**
Host TCP :8099 · netclaw-alert-receiver.service
POST /webhook → automatic-alert admission path
Enrich from Nautobot / inventory + RAG prior cases
Create / update initial case; send received notice
Other endpoints: /reinvestigate and /snapshot

Initial POST /api/events

Enrich + evaluate

**Automatic investigation policy**
Default T0: record / notify; no LLM investigation
T2 allowlist:
`WanHardDown`
`SwitchLinkLost [severity=critical]`
`DeviceSnmpExporterDown`
3 T2 admissions / hour · 30-minute duplicate TTL
Concurrency 2 applies to receiver trigger handling
Suppressed / over budget → diary; no backlog queue

Receiver /reinvestigate

03 POST /hooks/alert (T2)

**OpenClaw Gateway**
Host TCP :18789 · openclaw-gateway.service
main / HUD chat: GPT-6.1 Sol
alert / orchestrator: GPT-6 Luna
`/hooks/alert → agent alert · deliver=false`
main + alert share the Border OAuth credential store

04 n2n_route / n2n_task_wait via n2n-mcp

**iN2N Border / task broker**
netclaw-mesh.service · loopback TCP :11790
n2n-mcp is the Gateway's tool-facing client
Route → task ID → wait/progress → result

Chat / model controls → OpenClaw

05 iN2N: task dispatch / progress / result

## GUARDIAN EXECUTION PROFILE

Host worker; separate state, credentials and scoped tools

**Guardian member daemon**
netclaw-member-byrns-risk-guardian-claw.service
Python iN2N worker; persistent Border connection
`Identity: byrns-risk/guardian-claw`
Idle daemon ≠ continuously running LLM

Spawn model run per delegated task

**Scoped OpenClaw agent run**
GPT-6.1 Sol · GPT-6 Luna fallback
`openclaw agent --local --model <member-model>`
Existing OpenClaw embedded tool runtime
Own OAuth store / sessions / member environment
No separate Guardian Gateway listener

MCP tool calls / results

**Eight scoped MCP integrations**
prometheus-mcp · grafana-mcp · pyats-mcp
pfsense-mcp · unifi-network
greynoise-community-mcp · rag-mcp · memory-mcp
Read evidence; compare prior cases; assess cause
Investigation procedure: report, do not auto-remediate

Evidence → assessment

**Investigation outcomes**
Findings + evidence + root-cause classification
PATCH original case → resolved or escalated
Return task findings to the Border
Discord findings via configured notification path
POST /snapshot → RAG for future investigations

Device APIs / CLI reads through MCP

Need More + notes

06 PATCH same event ID: findings, root cause, status

07 Grafana queries via scoped MCP

Direct Prometheus MCP reads

Luna alert / Sol chat

Sol worker / Luna fallback

**OpenAI — ChatGPT / Codex subscription**
Same account allowance; separate Border / Guardian OAuth stores
Subscription inference; no general OpenAI API key required
Model calls are outbound; no network device changes implied

Telemetry / observation · Automatic investigation · Read / query path · Case lifecycle / operator feedback · Model inference

READING NOTES

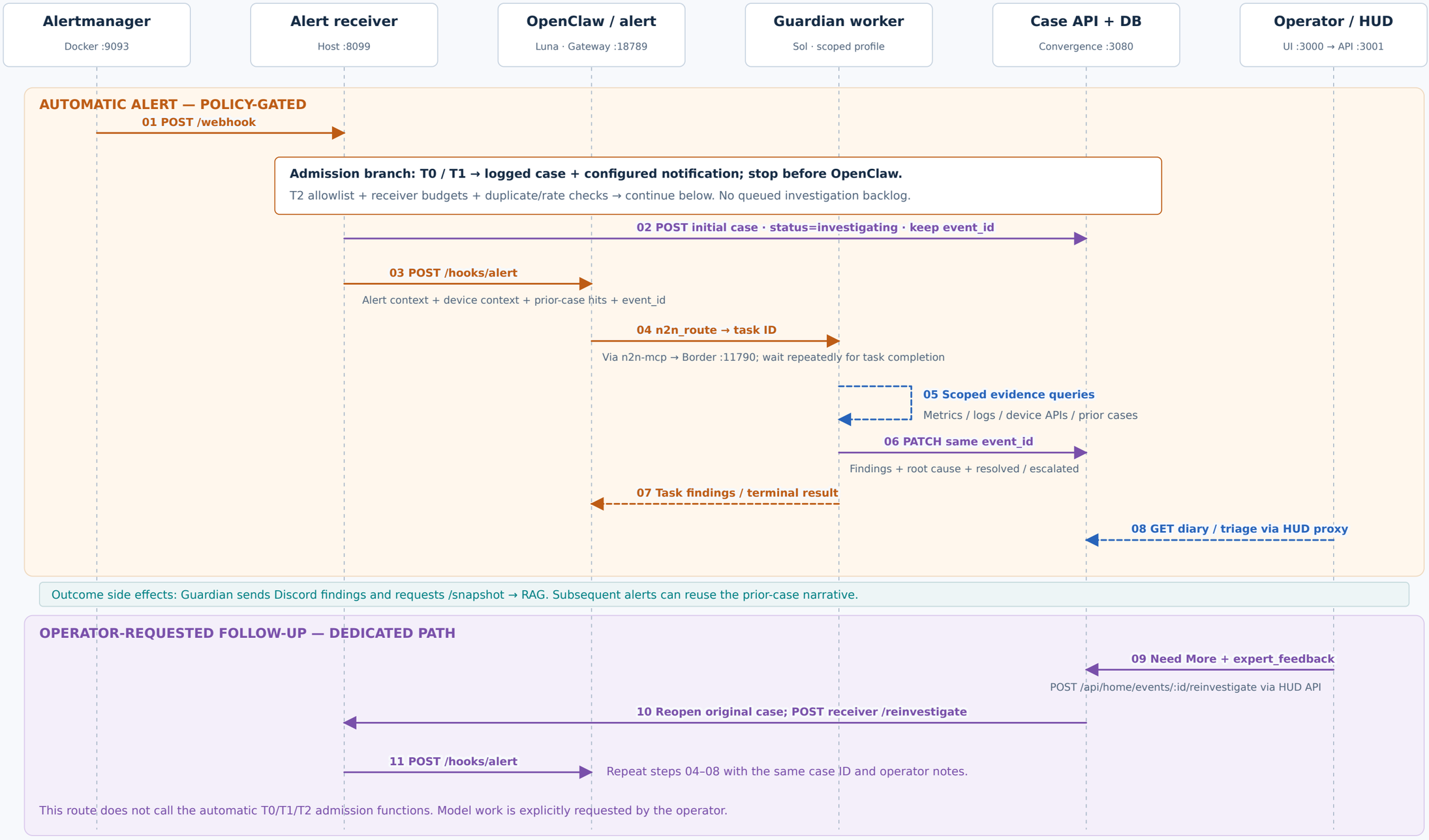
Need More uses /reinvestigate → /hooks/alert directly; automatic admission gates are not reapplied on that path.

Ports are host bindings unless marked internal/container. Dashed = query / feedback; dotted = model/control. Crossings are not junctions.

# GUARDIAN / CONVERGENCE — INVESTIGATION SEQUENCE

Automatic admission, delegated execution and the operator feedback loop • Current configured workflow






**IDENTITY + AUTHENTICATION**

Fingerprint groups the alert; event_id identifies the durable case; task_id identifies a delegated run; hook session key is fingerprint-based.
Upstream case-API bearer stays in the HUD proxy; hook bearer, iN2N member authentication and OAuth model auth are distinct.

Current receiver concurrency accounting ends after hook dispatch; it is not an end-to-end reservation for the worker's entire run.
Deployment bindings / model replies / member enrollment verified. Agent tool use, case completion and notifications follow the configured procedure.